

# MD RAKIBUL ISLAM RAIHAN

## ML Systems Engineer | LLM Inference & Multimodal AI

2027 New Grad · Available July 2027 · Open to Relocation · Xi'an, China  
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GitHub: github.com/theraihanrakibb · LinkedIn: linkedin.com/in/theraihanrakib · Portfolio: theraihanrakibb.github.io/Online-Portfolio

### SUMMARY

ML Systems Engineer focused on LLM inference, GPU optimization, and multimodal generative AI. At InfiX.ai, optimized production LLM serving - Qwen on 4× A800 and DeepSeek on multi-node H200 clusters - reducing long-context TTFT 8.3× (55s→6.7s) via 3-tier KV-cache offload and raising stable concurrency 192→256 via RDMA-based prefill/decode disaggregation. First author of an IEEE TNNLS manuscript on NVFP4 4-bit audio-guided video diffusion (4× memory reduction on H100). Strong in SGLang, CUDA/Triton, low-precision inference, and distributed systems.

### EXPERIENCE

#### AI Infrastructure Engineer Intern, InfiX.ai, Shenzhen, China, Apr 2026 - Jun 2026

- Reduced long-context TTFT 8.3× (55s→6.7s) and sustained a **92%** cache hit rate by designing and deploying a 3-tier KV-cache offload system (GPU HBM → CPU DRAM → SSD) over Mooncake RDMA for Qwen3.5/3.6-27B on 4× A800-80GB (32K+ token contexts).
- Improved stable serving concurrency 33% (192→256 requests) by implementing 2+2 GPU prefill/decode disaggregation with RDMA-based KV transfer, RadixAttention prefix caching, and continuous batching for Qwen3.6-27B.
- Modified **SGLang** cache-management and memory-allocation internals (kvcacheio.py, memory\_pool\_host.py, hi\_mamba\_radix\_cache.py) - across **v0.5.9** (A800) and **v0.5.10-post2** container image (H200) - to resolve production cache failures and support hybrid-attention (Mamba + full-attention) Qwen models.
- Profiled CUDA/Triton inference with **Nsight Systems**: GEMM accounted for **60%** of step time, redirecting optimization from cache I/O to compute kernels; FP8 KV cache halved per-token KV memory (72.9 → 36.5 KB).
- Deployed Qwen3-235B-A22B-FP8 (235B MoE) on 2× H200 and benchmarked DeepSeek-V3.2 (FP8) on 8× H200 - **91 ms TTFT / 36 ms TPOT** at 32-64 concurrency, **13.16 req/s** at 256 concurrency.

#### Software Engineer Intern (AI Agent), Hong Kong Amraim Group Co., Ltd., Shenzhen, China, Jun 2025 - Sep 2025

- Architected and delivered two end-to-end full-stack applications from concept to production deployment, using React front-ends and RESTful back-end services.
- Engineered AI agents that automated key marketing workflows, reducing manual effort and boosting team productivity by **25%**.

#### Electrical Software Engineer Intern, Shaanxi Longong Intelligent Technology Co., Ltd., Xi'an, China, Feb 2025 - Apr 2025

- Engineered real-time motor-control software on CODESYS PLCs using Modbus/CAN/Ethernet-IP protocols, improving system response time by **15%** to IEC 61131 industrial safety standards.

### EDUCATION

**M.Eng. in Software Engineering**, School of Software, Northwestern Polytechnical University - Sep 2024 - Jul 2027

GPA: 88/100, Top 1%, Chinese Government Scholarship, President of International Student Union (ISU)

**B.Eng. in Computer Science & Technology**, School of Computer Science, Northwestern Polytechnical University - Sep 2020 - Jul 2024

GPA: 85/100, Top 1%, Presidential Scholarship, Best Undergraduate Thesis Award

### RESEARCH

#### Graduate Researcher, Northwestern Polytechnical University, Sep 2024 - Present

Advisor: Prof. Zhang Peng

- Developed **NVFP4-DiT**, a **4-bit audio-guided video diffusion transformer**, with custom FP4-packed **Triton** kernels that are **3.2× faster than FP16** (2.1× over naive FP4) and cut memory **4×** at competitive generation quality.
- Implemented adaptive block-wise **FP8/FP4/NVFP4** quantization and quantization-aware training with learnable per-head cross-modal scales; evaluated trade-offs on WebVid-10M, VGGSound, and UCF-101.
- Built a **vLLM-style serving scheduler** (frame-level paged attention, dynamic frame batching) and benchmarked on **H100 GPUs** using PyTorch Profiler and Nsight to guide low-precision kernel choices.
- Research on multimodal deepfake detection using visual, audio, and temporal consistency analysis.
- M.Eng. Thesis: "Detecting Deepfake Video by a Multimodal Audio-Visual Framework with Temporal Inconsistencies."

#### Undergraduate Researcher, Northwestern Polytechnical University, Jan 2023 - Jul 2024

Advisor: Dr. Yaxing Chen

- Designed and built a distributed confidential-query framework on Apache Spark + CKKS homomorphic encryption, validated on a **10-node cluster**: **95% privacy preservation** and **20% performance improvement** via optimized analytics pipelines.
- B.Eng. Thesis: "Design and Implementation of a Distributed Confidential Query Protocol for Apache Spark."

### PROJECTS

#### AI Infrastructure Portfolio, github.com/theraihanrakibb/ai-infra-portfolio

- Designed and shipped 10 Dockerized AI-infra services - LLM gateway, model serving, GPU observability, KV-cache simulation, benchmarking, quantization, tracing, and RAG - each with CI and a reproducible quickstart.

#### Distributed Training Launcher, github.com/theraihanrakibb/train-launcher

- Built a PyTorch distributed CLI that launches and supervises multi-node training jobs with fault detection, automatic retry, and checkpoint recovery.

### TECHNICAL SKILLS

**LLM Inference & Systems**: SGLang, vLLM, KV Cache, PD Disaggregation, Mooncake/RDMA, Continuous Batching, RadixAttention

**GPU & Performance**: CUDA, CUDA C++, Triton, Nsight Systems, Nsight Compute, PyTorch Profiler, GEMM optimization, FP8/FP4/INT4

**ML & Multimodal**: PyTorch, Hugging Face Transformers, Diffusion/DiT, Multimodal Learning, OpenCV

**Distributed Systems**: PyTorch Distributed, Docker, Kubernetes, Linux, Prometheus, CI/CD

**Programming**: Python, C++, CUDA C++, Bash, SQL, Data Structures & Algorithms

### PUBLICATIONS & HONORS

**First Author** — "NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation" — IEEE Transactions on Neural Networks and Learning Systems (TNNLS, Q1), manuscript under review.

**Honors**: Chinese Government Scholarship (2024-2027); Presidential Scholarship (2020-2024); Wu Yajun Scholarship (2024); Best Undergraduate Thesis Award (2024); Outstanding Graduate (2024), NWPU; Hackathon Winner - NWPU AI Challenge (2023).

**Certifications**: AWS Certified Machine Learning – Specialty; Professional Scrum Master I (PSM I), Scrum.org; Deep Learning Specialization, DeepLearning.AI.

# MD RAKIBUL ISLAM RAIHAN

## ML 系统工程师 | 大模型推理与多模态 AI

2027 届应届 · 2027 年 7 月到岗 · 可接受异地 · 中国·西安

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### 个人简历

ML 系统工程师, 专注大模型推理、GPU 优化与多模态生成式 AI。在 InfiX.ai 实习期间, 优化生产级大模型服务——Qwen 部署于 4× A800、DeepSeek 部署于 H200 多节点集群——通过三级 KV-cache 卸载将长上下文 TTFT 降低 8.3× (55s→6.7s), 并借助基于 RDMA 的 prefill/decode 分离将稳定并发由 192→256 提升。作为第一作者撰写 IEEE TNNLS 论文, 研究 NVFP4 4 比特音频引导视频扩散, 在 H100 上实现显存降低 4×。擅长 SGLang、CUDA/Triton、低精度推理与分布式系统。

### 工作经历

#### 无界智索 (深圳) 科技 (InfiX.ai), AI 基础设施工程师实习, 深圳, 2026年4月 - 2026年6月

- 在 4× A800-80GB 上基于 Mooncake RDMA 设计并实现三级 KV-cache 卸载系统 (GPU HBM → CPU DRAM → SSD), 服务 Qwen3.5/3.6-27B 时将长上下文 (32K+ token) 首字延迟 (TTFT) 降低 8.3× (55s→6.7s), 缓存命中率维持 **92%**。
- 通过基于 RDMA 的 KV 传输实现 2+2 GPU prefill/decode 分离, 结合 RadixAttention 前缀缓存与连续批处理, 将 Qwen3.6-27B 的稳定服务并发提升 33% (192→256 请求)。
- 修改 **SGLang** 缓存管理与内存分配内部模块 (kvcacheio.py、memory\_pool host.py、hi mamba radix\_cache.py) ——覆盖 **v0.5.9** (A800) 与 **v0.5.10-post2** 容器镜像 (H200) ——修复生产缓存故障并支持混合注意力 (Mamba + 全注意力) Qwen 模型。
- 使用 **Nsight Systems** 分析 CUDA/Triton 推理: GEMM 占单步耗时 **60%**, 由此将优化重心从缓存 I/O 转向计算算子; FP8 KV cache 将单 token KV 显存减半 (72.9 → 36.5 KB)。
- 在 2× H200 部署 Qwen3-235B-A22B-FP8 (235B MoE), 并在 8× H200 上对 DeepSeek-V3.2 (FP8) 基准测试: **91 ms TTFT / 36 ms TPOT** (32-64 并发)、**13.16 req/s** (256 并发)。

#### 香港安莱姆集团有限公司 (Amraim Group Co., Ltd.), 软件工程师实习 (AI Agent), 深圳, 2025年6月 - 2025年9月

- 从构想到生产部署, 架构并交付两个端到端全栈应用 (React 前端 + RESTful 后端服务)。
- 开发 AI Agent 自动化关键营销 workflow, 减少人工工作量, 团队生产力提升 **25%**。

#### 陕西龙宫智能科技有限公司 (Longong), 电气软件工程师实习, 西安, 2025年2月 - 2025年4月

- 基于 CODESYS 开发 PLC 实时电机控制软件, 使用 Modbus、CAN 与 Ethernet/IP 协议, 系统响应时间提升 **15%**, 符合 IEC 61131 工业安全标准。

### 教育背景

#### 软件工程硕士, 软件学院, 西北工业大学 (985/211) - 2024年9月 - 2027年7月

GPA: 88/100, 前 1%, 中国政府奖学金, 国际学生联合会主席 (ISU)

#### 计算机科学与技术学士, 计算机学院, 西北工业大学 (985/211) - 2020年9月 - 2024年7月

GPA: 85/100, 前 1%, 校长奖学金, 最佳本科毕业论文奖

### 研究经历

#### 研究生研究员, 西北工业大学, 2024年9月 - 至今

导师: 张蓬 教授

- 开发 **NVFP4-DiT: 4 比特音频引导视频扩散 Transformer**, 自研 FP4 打包 **Triton** 算子相较 FP16 提速 **3.2 倍** (相较朴素 FP4 提速 2.1 倍), 显存降低 **4 倍**, 生成质量具有竞争力。
- 实现自适应分块 **FP8/FP4/NVFP4** 量化与量化感知训练 (跨模态注意力可学习 per-head 缩放), 并在 WebVid-10M、VGGSound 与 UCF-101 上评估质量-效率权衡。
- 构建 **vLLM 风格的服务调度器** (帧级分页注意力、动态帧批处理), 并在 **H100** 上使用 PyTorch Profiler 与 Nsight 进行基准测试, 指导低精度算子选型。
- 基于视觉、音频与时序一致性的多模态深度伪造检测研究。
- 硕士论文: 《检测深度伪造视频: 基于多模态音视频框架与时序不一致性》。

#### 本科研究员, 西北工业大学, 2023年1月 - 2024年7月

导师: 陈亚星 博士

- 基于 Apache Spark 与 CKKS 同态加密设计并实现分布式机密查询框架, 在 **10 节点集群**上验证, 通过优化分析管线实现**95% 隐私保护与20% 性能提升**。
- 本科论文: 《面向 Apache Spark 的分布式机密查询协议设计与实现》。

### 项目

#### AI 基础设施代码集, [github.com/theraihanrakibb/ai-infrastructure](https://github.com/theraihanrakibb/ai-infrastructure)

- 设计并交付 10 个 Docker 化的 AI 基础设施服务——LLM 网关、模型服务、GPU 可观测性、KV-cache 模拟、基准测试、量化、链路追踪与 RAG, 均配备 CI 与可复现的上手流程。

#### 分布式训练启动器, [github.com/theraihanrakibb/train-launcher](https://github.com/theraihanrakibb/train-launcher)

- 开发 PyTorch 分布式命令行工具, 支持多节点训练任务的启动与监控, 并具备故障检测、自动重试与断点恢复。

### 技术技能

大模型推理与系统: SGLang, vLLM, KV Cache, PD 分离, Mooncake/RDMA, 连续批处理, RadixAttention

GPU 与性能: CUDA, CUDA C++, Triton, Nsight Systems, Nsight Compute, PyTorch Profiler, GEMM 优化, FP8/FP4/INT4

机器学习与多模态: PyTorch, Hugging Face Transformers, Diffusion/DiT, 多模态学习, OpenCV

分布式系统: PyTorch Distributed, Docker, Kubernetes, Linux, Prometheus, CI/CD

编程语言: Python, C++, CUDA C++, Bash, SQL, 数据结构与算法

### 论文与荣誉

第一作者 — 《NVFP4-DiT: 面向低成本视频生成的高效 4 比特音频引导视频扩散 Transformer》— IEEE TNNLS (Q1), 审稿中。

荣誉: 中国政府奖学金 (2024-2027); 校长奖学金 (2020-2024); 吴亚军奖学金 (2024); 最佳本科毕业论文奖 (2024); 西北工业大学优秀毕业生 (2024); NWPU AI 挑战赛黑客松冠军 (2023)。

认证: AWS 认证机器学习 — 专业级; Professional Scrum Master I (PSM I), Scrum.org; Deep Learning 专项课程, DeepLearning.AI。